Polarise Telecom Technology

System Architecture, Engineering Techniques, Future Roadmap & Maintenance Budget

Target Platform: Cloud PBX & Admissions CRM	**Production Host:** Vercel Serverless + Neon Cloud PostgreSQL
Client Organization: Polaris Educational Consultancy	**Live URL:** https://polarise-telecom.vercel.app
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1. Executive Summary & System Mission

Polarise Telecom Technology is a cloud-native telecalling infrastructure and admissions CRM engineered to solve the operational challenges of student outreach, parent counseling, and lead disposition. The platform bridges counseling teams with parents through a unified virtual institutional identity (+91 80471 90000), shielding personal mobile numbers while recording, transcribing, and auditing 100% of outbound conversations.

Key operational milestones achieved in the current deployment include: persistent PostgreSQL cloud persistence, role-based security boundaries, zero-reload SPA navigation, instant click-to-call dialing, real-time live monitoring, and dual-mode telephony bridging (Zero-Cost Synthetic Sandbox and Live Exotel Carrier Trunk).

2. Current Engineering Techniques & Architecture

2.1 Frontend Engineering Patterns

Pattern / Technique	Stack	Engineering Implementation & Benefit
Reactive Component Tree	React 18, TypeScript	Strongly typed components eliminate runtime null errors and guarantee responsive state synchronization across counseling suites.
Chroma Waves Engine	HTML5 Canvas 2D	Multi-harmonic sinusoidal algorithm running at 60 FPS renders dynamic chromatic ambient light (amber, rose, violet, cyan) on login.
SPA Fallback Routing	React Router v6 + Vercel Rewrites	Client-side wildcard rewrite (/(.*) → /index.html) ensures zero 404 errors during direct link bookmarking, browser back/forward, and logout redirects.
Client-Side Phone Guard	JavaScript RegEx Engine	Normalizes lead phone numbers by stripping +91, 0, spaces, and hyphens into standard 10-digit ITU compliance before database transmission.
Tokenized Audio Streaming	HTML5 Audio API	Streams byte-range scrubbable WAV audio with query-token authentication strictly restricted to administrators for quality review.

2.2 Backend, Persistence & PBX Architecture

Technique / Component	Technology	Engineering Functionality
Async REST Gateway	Python 3.12, FastAPI, Starlette	High-concurrency ASGI serverless API delivering sub-50ms endpoint latencies with Pydantic v2 data validation schemas.
Cloud Relational Persistence	Neon Cloud PostgreSQL	Dedicated managed cloud database in AWS US-East-1 with connection pooling, preserving all leads, calls, sessions, and audits across ephemeral serverless lifecycles.
Dual-Driver Engine	psycopg 3 & psycopg2	SQLAlchemy 2.0 ORM with connection pre-ping validation eliminates stale sockets and handles burst telecaller concurrency seamlessly.
2-Leg Telephony Bridging	Exotel REST + Sandbox PBX	Dials agent leg first, confirms counselor pickup, dials student lead, and bridges audio streams with carrier-grade recording webhooks.
Synthetic Waveform Engine	Python wave & struct	Synthesizes realistic 8kHz mono WAV audio buffers dynamically in sandbox mode, enabling full zero-cost testing and compliance simulation.
Security & RBAC	JWT + Bcrypt (HS256)	12-hour encrypted bearer tokens, strict role segregation, and complete isolation of call recordings from counselor suite views.

3. Future Technical Roadmap (Techniques to Add)

To elevate Polarise Telecom from a telecalling CRM into an automated, AI-powered admissions engine, the following four high-impact technical modules are recommended for subsequent deployment phases:

Phase & Technique	Architecture & Protocol	Business & Cost Impact
PHASE 1 (Q1) WebRTC Softphone	WebRTC / SIP.js / WebSockets inside React frontend. Counselors dial via browser headset.	**Eliminates Leg-1 carrier call cost** to agent phones. Lowers telecom minute bills by ~45% and reduces call connection latency.
PHASE 2 (Q2) AI Speech-to-Text & Quality Scoring	Google Gemini 1.5 Flash / OpenAI Whisper audio pipeline triggered on recording webhook.	Automated transcription, sentiment detection, key topic extraction (courses, budget), and objective counselor pitch compliance scoring.
PHASE 3 (Q3) WhatsApp Business & DLT SMS Cloud Suite	Meta WhatsApp Cloud API + MSG91 DLT gateway integration.	Auto-dispatches course prospectus, college fee details, and follow-up links to parent WhatsApp immediately upon call disposition.
PHASE 4 (Q4) Predictive Auto-Dialer & Branch Partitioning	Redis queue & Celery background workers; automated lead pacing algorithms.	Triples counselor throughput from 60 calls/day to 180+ calls/day by auto-filtering busy/unreachable lines before handing to agents.

4. Maintenance Budget & Financial Projections

The serverless cloud micro-architecture minimizes upfront capital expenditure. Below is the itemized operational and infrastructure cost projection structured across three progressive deployment tiers:

Expense Category	Service Provider	Tier 1: Current Pilot (10 Counselors)	Tier 2: Expansion (25 Counselors)	Tier 3: Enterprise (50+ Counselors)
Frontend & API Hosting	Vercel Serverless	**Free ($0)**	Vercel Pro ($20/mo = ■1,660)	Vercel Pro+ ($40/mo = ■3,320)
Managed Cloud DB	Neon PostgreSQL	**Free ($0, <0.5GB)**	Launch ($19/mo = ■1,580)	Scale ($69/mo = ■5,730)
Virtual DID Trunk	Exotel / Tata Tele	■1,000 / mo	■2,000 / mo (2 DIDs)	■4,000 / mo (4 DIDs)
Voice Call Minutes	Carrier Pulse Rates	■3,000 – ■5,000 / mo (~10k mins @ 38p/min)	■8,000 – ■12,000 / mo (~25k mins)	■20,000 – ■28,000 / mo (~60k mins)
Recording Audio Vault	Cloudflare R2 / AWS S3	■150 / mo (~20GB)	■450 / mo (~60GB)	■1,200 / mo (~150GB)
Domain & SSL Cert	Namecheap / GoDaddy	■100 / mo (■1,200/yr)	■100 / mo (■1,200/yr)	■100 / mo (■1,200/yr)
DevOps & Bug Fixes	Software Maintenance	■3,000 – ■5,000 / mo	■6,000 – ■10,000 / mo	■12,000 – ■18,000 / mo
TOTAL MONTHLY	**All Inclusive**	**■4,100 – ■7,760 ($50 – $95 USD)**	**■14,000 – ■19,000 ($170 – $230 USD)**	**■35,000 – ■49,000 ($420 – $590 USD)**

4.2 Capital ROI: Traditional PBX vs. Polarise Telecom

Expenditure Metric	Legacy On-Premise PBX	Polarise Cloud Platform	Net Capital Savings
Hardware PBX Server	■1,50,000 – ■2,50,000	**■0 (Serverless Cloud)**	**100% Upfront Capital Saved**
Desk IP Hardware Phones	■35,000 (10 IP Phones)	**■0 (Counselor Browser/SIM)**	**■35,000 Saved**
Server AMC & PRI Line	■45,000 / year	**Included in Cloud SLA**	**■45,000 / year Saved**
Go-Live Deployment	3 to 5 Weeks	**Instant (Active Today)**	**Zero Onsite Delay**

5. Strategic Recommendations & Action Plan

1. Immediate Next Step (Month 1): Continue current live operations on the Vercel + Neon Cloud PostgreSQL tier. Plug in production Exotel API credentials whenever the consultancy decides to shift from Demo Sandbox to carrier billing.
2. Mid-Term Optimization (Months 2–4): Implement the in-browser WebRTC Softphone module to eliminate agent leg carrier charges, cutting recurring phone minute costs in half.
3. High-Value Automation (Months 5–6): Connect WhatsApp Cloud API for automated student prospectus dispatch and activate Gemini voice summaries to eliminate manual counselor disposition writing.